Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 20

Understand the K-means clustering algorithm for a given dataset

Experiment 20: Understand the K-means clustering algorithm for a given dataset

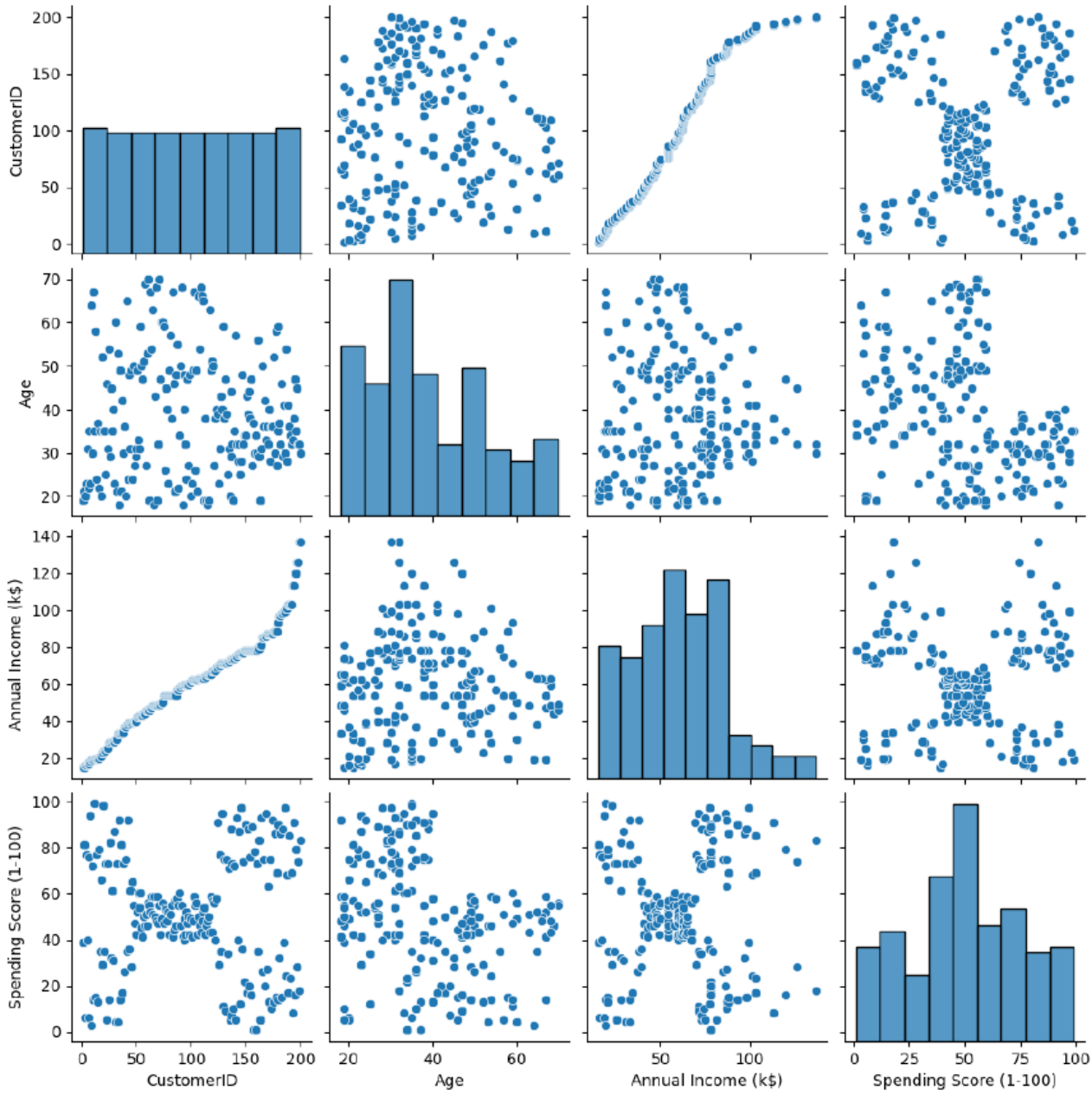
Aim

To apply the K-means clustering algorithm to the given dataset.

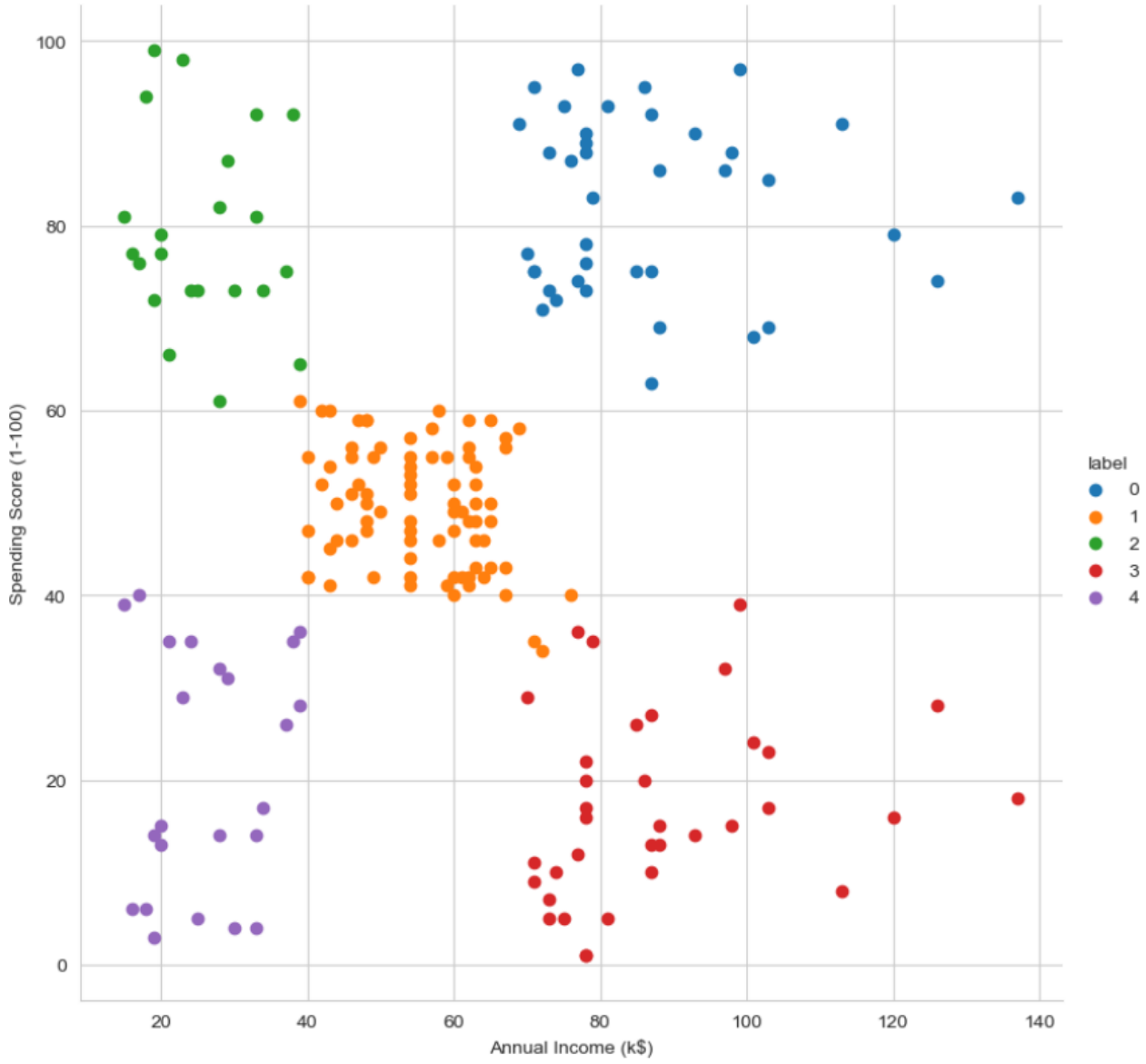
Source Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
df=pd.read_csv('Mall_Customers.csv')
df.info()
df.head()
sns.pairplot(df)
features=df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
Final['label']=model.predict(features)
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

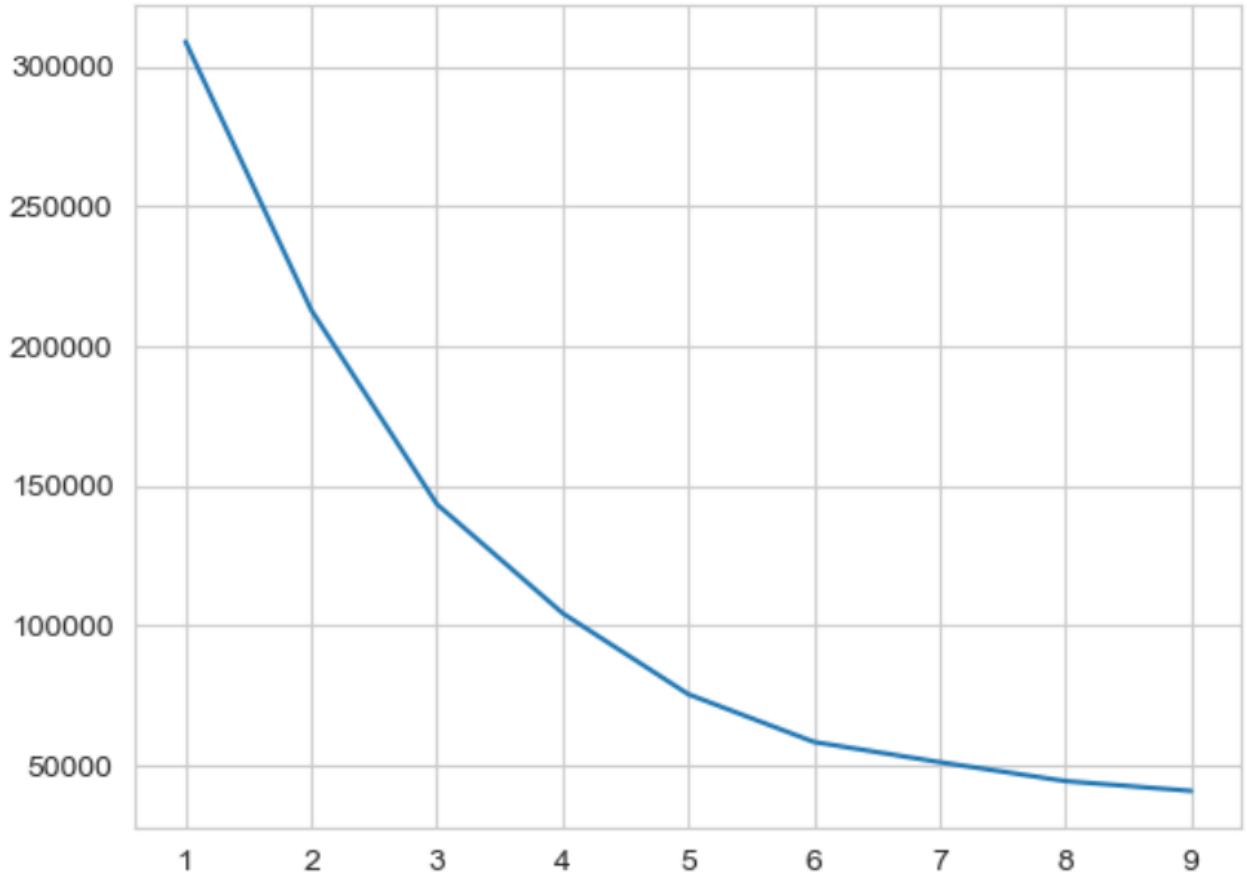
Output - Experiment 20



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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
CustomerID Gender Age Annual Income (k$) Spending Score (1-100)
0 1 Male 19 15 39
1 2 Male 21 15 81
2 3 Female 20 16 6
3 4 Female 23 16 77
4 5 Female 31 17 40
KMeans(n_clusters=5)
Annual Income (k$) Spending Score (1-100) label
0 15 39 4
1 15 81 2
2 16 6 4
3 16 77 2
4 17 40 4

Result - Experiment 20

K-means clustering was applied to the given dataset.